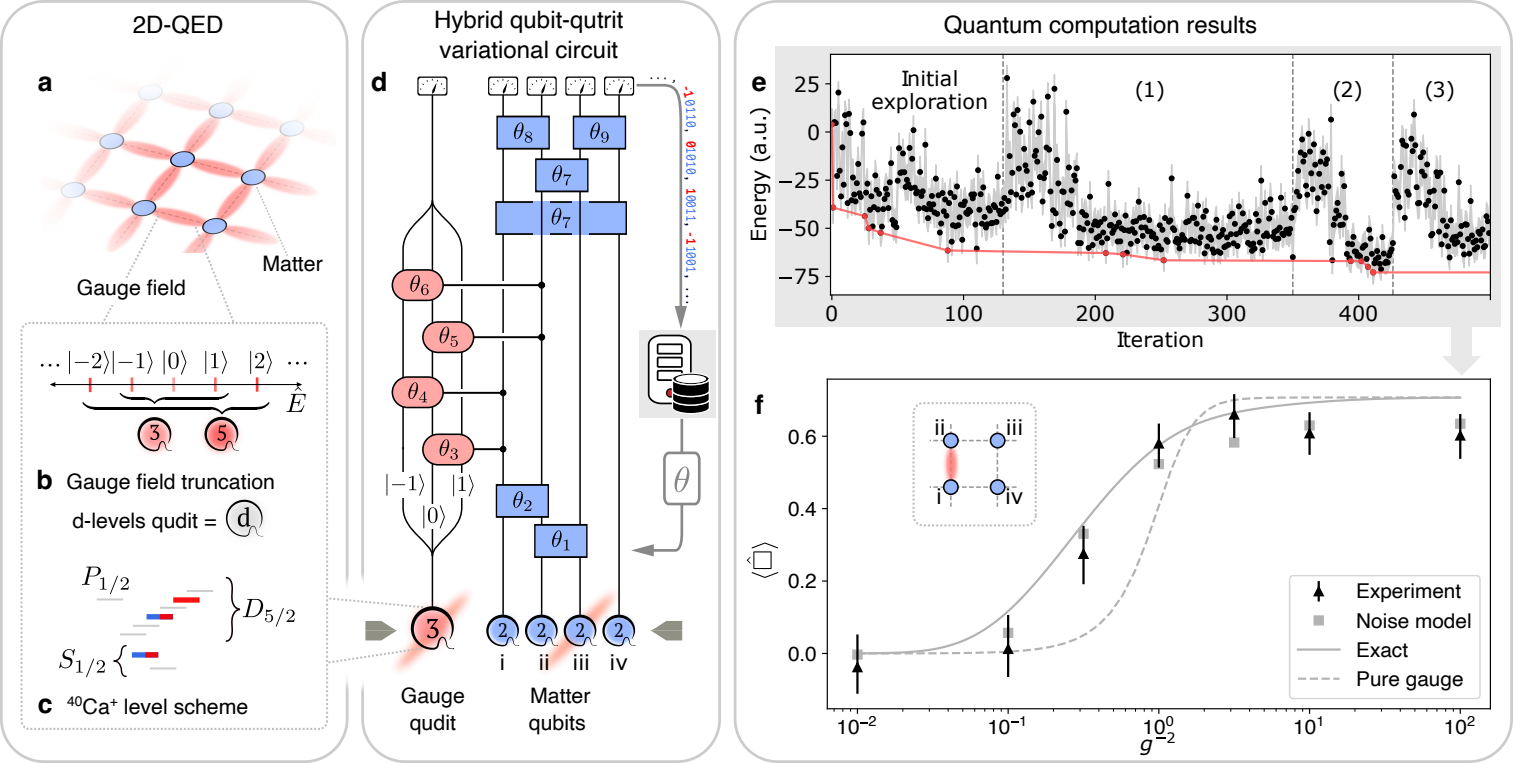

2D-QED
a
Gauge field
Matter
$\cdots |-2\rangle |-1\rangle |0\rangle |1\rangle |2\rangle \cdots$
$\vec{E}$
3
5
b Gauge field truncation
d-levels qudit = d
$P_{1/2}$
$D_{5/2}$
$S_{1/2}$
c $^{40}Ca^+$ level scheme
Hybrid qubit-qutrit variational circuit
d
θ_8
θ_9
θ_7
θ_7
θ_6
θ_5
θ_4
θ_3
θ_2
θ_1
$|-1\rangle$
$|1\rangle$
$|0\rangle$
-10110, 01010, 10011, -11001, ...
θ
3
2
2
2
2
i
ii
iii
iv
Gauge qudit
Matter qubits
Quantum computation results
e
Energy (a.u.)
Initial exploration
(1)
(2)
(3)
Iteration
f
$\langle \hat{\square} \rangle$
ii
iii
i
iv
Experiment
Noise model
Exact
Pure gauge
g^2